

Muhammad Junayed

CUET, Raozan, Chattogram, Bangladesh | mdjunayed573@gmail.com | +880 1876220119 | LinkedIn | GitHub

EDUCATION

Bachelor of Science in Electronics and Telecommunication Engineering

March 2022–Present

Chittagong University of Engineering and Technology

- CGPA: 3.51/4.00 (upto 7th semester)

WORK EXPERIENCES

PORIDHI.IO

May 2025 – November 2025

Software Engineer Intern

- Worked on software engineering, cloud computing and backend development tasks during internship tenure.

BRAIN STATION 23 PLC

May 2025

Industrial Trainee

- Completed industrial training focused on modern software engineering workflows and enterprise systems.

RESEARCH & PUBLICATIONS

B.Sc. Thesis:

- **Know-When-You-Don't-Know: Detecting and Mitigating Hallucinations in Large Language Models through Internal-State Awareness and Closed-Loop RAG Systems.**

Supervisor: Priyonti Paul Tumpa, Assistant Professor, ETE, CUET

Conference Publications:

- **Muhammad Junayed, Tanzeem Tahmeed Reza, Md. Saiful Islam. Enhancing Defect Recognition: Convolutional Neural Networks for Silicon Wafer Map Analysis.** In: 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), Gazipur, Bangladesh. IEEE, 2024. DOI: 10.1109/ICAEEE62219.2024.10561853.
- **Muhammad Junayed, Tanzeem Tahmeed Reza. Leveraging Vision Transformer to Classify Cancer Stages from Breast Ultrasound Images.** In: IEEE International Conference on Signal Processing, Information, Communication and Systems 2025 (SPICSCON), Rajshahi University, Bangladesh, 2025. DOI: 10.1109/SPICSCON69221.2025.11504239.
- Md. Abdur Rahman, Md. Al Amin, Sabik Aftahee, **Muhammad Junayed**, Md. Ashiqur Rahman. **SmolLab_SEU at BEA 2025 Shared Task: A Transformer-Based Framework for Multi-Track Pedagogical Evaluation of AI-Powered Tutors.** In: Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025). Association for Computational Linguistics (ACL), 2025, pp. 1127–1134.

PROJECTS

AroBot – Agentic RAG Multi-Modal Chatbot System

- Built a production-ready medical assistant using Llama 3.2:3B and LLaVA:7B via Ollama, with Pinecone vector search and FastAPI deployment.
- Designed modular agentic workflows for text, image, and PDF processing using OCR and RAG pipelines.
- Implemented prescription OCR, CLIP vision pipeline, LangSmith tracing, and DuckDuckGo search integration.
- Added conversation memory via Model Context Protocol, rate-limited APIs, and a responsive chat UI so clinicians get auditable session histories, health metrics, and ingestion scripts for continuously updating vector indexes.

Uber Fare Prediction with MLOps Orchestration

- Built an end-to-end MLOps pipeline using Airflow for scheduling, ZenML for pipeline orchestration, and Flask for deployment.
- Used MLflow for experiment tracking and selection of optimal ensemble models for production.
- Integrated PostgreSQL for metadata storage and system-level logging to support reproducibility and monitoring.

Bangladesh Medicine Scrape

- Built a Django + Scrapy + Playwright pipeline that reuses verified browser cookies to crawl *MedEx* medicine data.
- Stored results in PostgreSQL and exposed them via Django Admin and token-authenticated REST APIs (DRF).

NodeScape – Graph Algorithm Visualizer + Graph Type Predictor

- Built a full-stack React + Flask/TensorFlow platform for graph visualization and graph-type prediction using LightGIN GNN.
- Implemented step-by-step BFS/DFS visualization and Docker deployment support.

Face Recognition & RFID Smart Attendance System

- Developed a Dockerized smart attendance system on Jetson Nano integrating MTCNN for face detection and ResNet-18 for face recognition, enabling real-time identity verification.
- Integrated JT308 RFID module and Canon EOS 250D camera for multi-modal attendance tracking, improving robustness and reducing spoofing in real-world deployment.

Credit Card Fraud Detection

- Built ML models using Random Forest, XGBoost, and LightGBM on highly imbalanced transaction datasets.
- Achieved approximately 0.98 AUPRC and 0.96 F1-score with Streamlit deployment.

DataBench Question Answering System (SemEval 2024 Task 8)

- Developed an AI-powered Question Answering (QA) system for structured tabular data, integrating rule-based extraction, semantic similarity matching, and transformer-based NLP models (BERT-large-uncased-squad2) for accurate data-driven responses.

EEG Data Analysis & Alcoholism Detection

- Analyzed EEG sensor data and developed KNN and SVM models achieving up to 99% accuracy.

Asynchronous Task Processing System (Flask + Celery + AWS Pulumi)

- Built distributed asynchronous task processing pipelines using Flask, Celery, RabbitMQ, Redis, Docker, and AWS EC2.
- Automated infrastructure provisioning using Pulumi (IaC) and implemented both single-node and multi-node EC2 deployment.

RickshawX – Smart Mobility for CUET (Microservices Ride-Sharing Platform)

- Built a full-stack ride-sharing platform for CUET using a scalable **microservices architecture** with Node.js/Express, React, MongoDB, Redis, RabbitMQ, Docker, and Nginx.
- Developed independently deployable Auth, Ride, Trip, Payment, and Notification services with JWT authentication, centralized API Gateway routing, and asynchronous inter-service communication.
- Implemented an **event-driven system** using RabbitMQ, GPS-based fare calculation with the Haversine formula, Redis caching, trip management, ratings, and real-time notifications.
- Containerized the complete platform with Docker Compose for production-style deployment and service orchestration.

Node.js Asynchronous Task Processing System

- Developed scalable async architecture using Node.js, RabbitMQ (job queueing), Redis (pub/sub updates), Socket.IO (real-time browser feedback), Docker Compose, and Pulumi ensuring live task tracking, failure recovery (DLQ), and system scaling via `–scale worker=n`

Home Power Automation System

- Built an ESP32-based IoT prepaid energy monitoring and remote AC load control system using MQTT and Firebase.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, PHP, Node.js, HTML, CSS, SQL
- **Frameworks & Tools:** Flask, FastAPI, React, Django, Docker, RabbitMQ, Redis, Pulumi, Numpy, Pandas, Matplotlib, Scikit learn, TensorFlow, Keras, OpenCV.
- **MLOps & Data Science:** AWS fundamentals, Container Networking, PostgreSQL, BigQuery, Power BI, MongoDB
- **Software:** MATLAB, VSCode, Cisco Packet Tracer, Adobe Photoshop, DaVinci Resolve
- **Typography:** LaTeX, MS Word, MS PowerPoint.

EXTRACURRICULAR ACTIVITIES

Televerse 1.0, CUET

October 2025–21th November 2025

Team Lead & Organizer

- Led Bangladesh's first-ever Shark Tank-style hardware-software integrated automation competition *IOTrix*.
- Coordinated interdisciplinary teams working on IoT, automation, and embedded system prototypes.

CUET Photographic Society (CUETPS)

23th September 2024–11th March 2026

Assistant Design Secretary

- Contributed to branding, visual design, and media content creation for organizational campaigns and events.

3rd International Conference on Electrical, Computer and Communication Engineering (ECCE), CUET

23th Februaary–25th Februaary 2023

Event Manager

- Managed event coordination, logistics, and execution for an international engineering conference hosted at CUET.

AWARDS AND CERTIFICATIONS

- **Champion** – ETE Infixon Case Solving Competition, CUET
- **1st Runner-up** – Programming Hackathon, Department of ETE, CUET
- **Finalist** – Revealing The Unseen: Season VI by CUET Photographic Society-CUETPS
- **13th at Datathon**, (out of 108 teams), BitFest, Dept. of CSE, KUET, Bangladesh (2025)
- **Finalist** – SemEval 2024 Task 8: DataBench
- **3rd Place** – Techathon: The Heart of Robolution 1.0 by by the Department of Mechatronics and Industrial Engineering, CUET.
- **Finalist** – BEA 2025 Shared Task on Pedagogical Evaluation of AI Tutors,organized by ACL SIGEDU; secured Top 10 in Tutor Identity (F1: 0.8621) and competitive ranks in 4 additional tracks
- **Honorable Mention** – Break The Monolith Microservice Hackathon by IEEE Computer Society, CUET